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Large Language Model Threats Taxonomy

AI Controls Framework Working group

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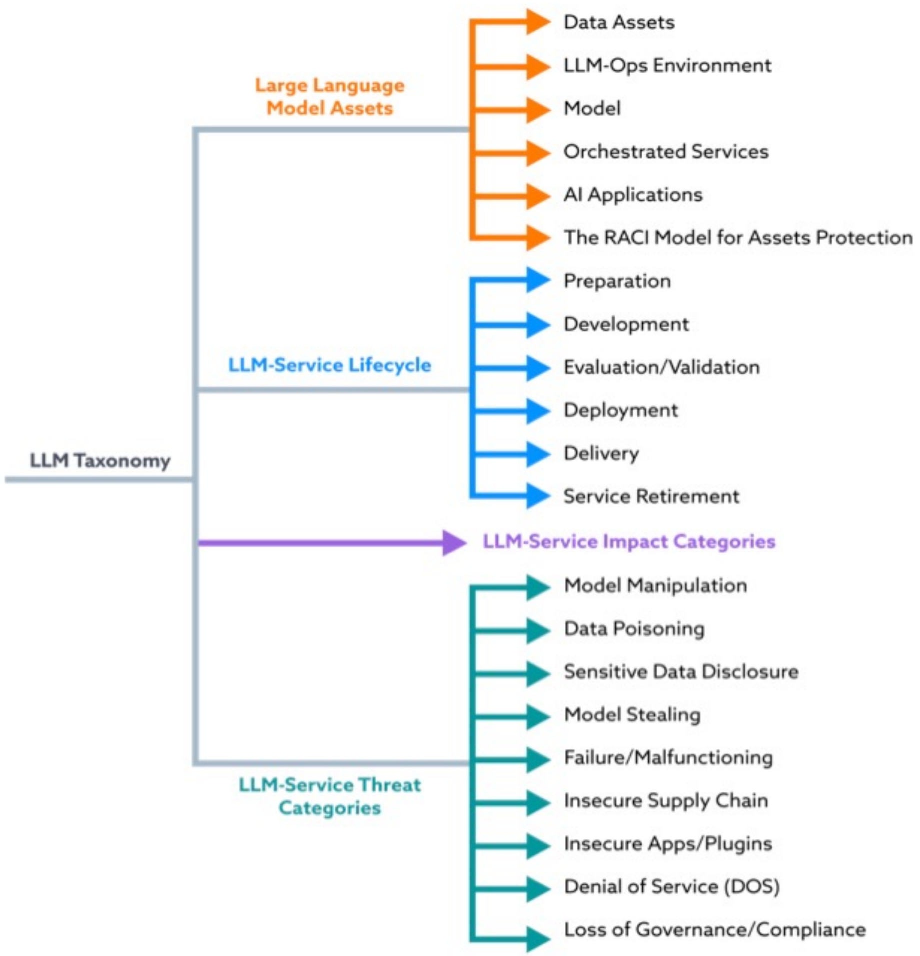
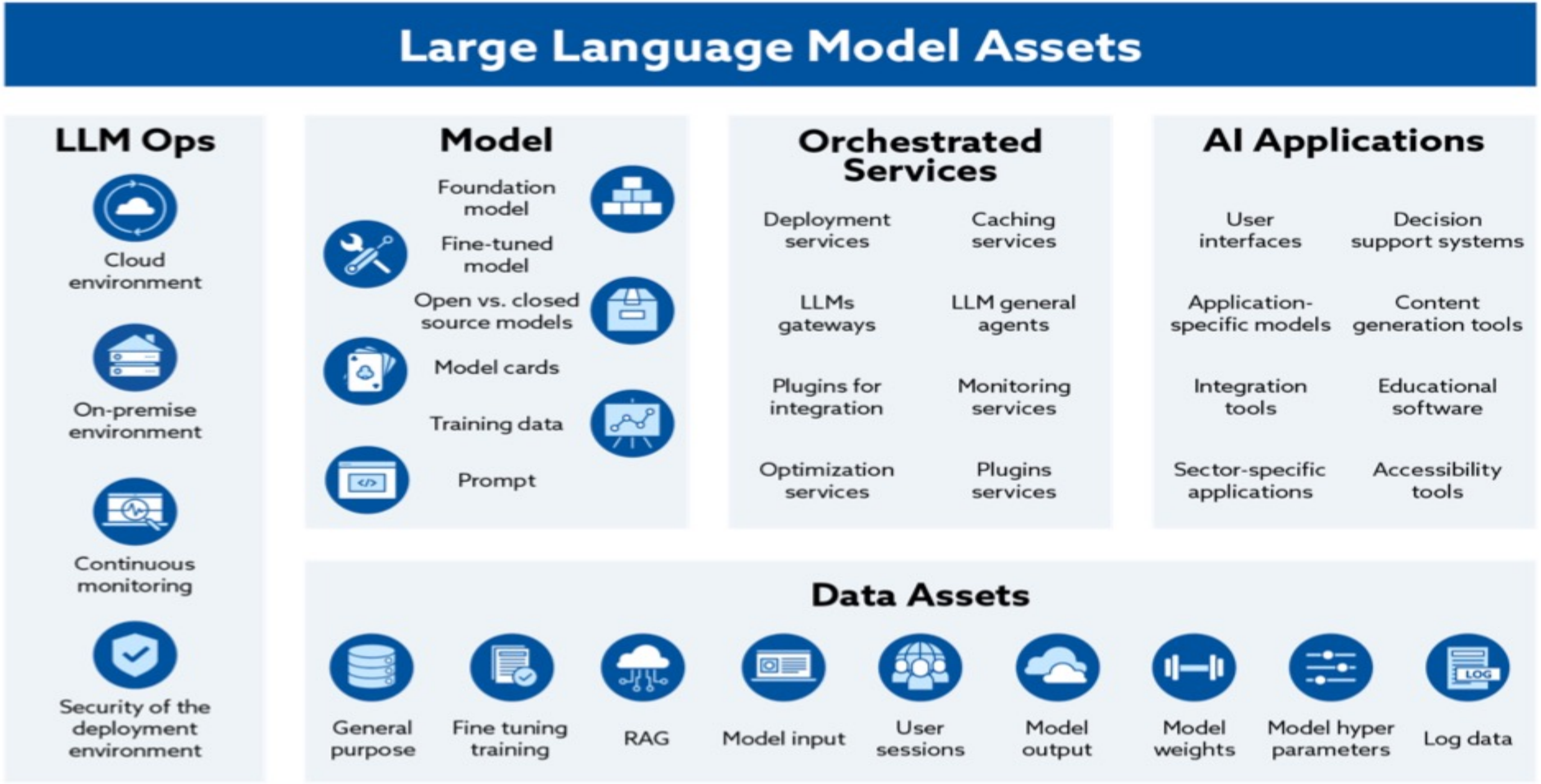
Cloud Security Alliance

Defining

- LLM Assets
- LLM-Service Lifecycle
- LLM-Service Impact Categories
- LLM-Service Threat Categories



LLM Assets



Large Language Model Assets
LLM Ops
Cloud environment
On-premise environment
Continuous monitoring
Security of the deployment environment
Model
Foundation model
Fine-tuned model
Open vs. closed source models
Model cards
Training data
Prompt
Orchestrated Services
Deployment services
Caching services
LLMs gateways
LLM general agents
Plugins for integration
Monitoring services
Optimization services
Plugins services
AI Applications
User interfaces
Decision support systems
Application-specific models
Content generation tools
Integration tools
Educational software
Sector-specific applications
Accessibility tools
Data Assets
General purpose
Fine tuning training
RAG
Model input
User sessions
Model output
Model weights
Model hyper parameters
Log data

LLM-Ops Cloud Environment

1. Cloud running the training environment
2. Cloud running the model inference point
3. Cloud running the AI applications
4. Hybrid and multi-cloud infrastructure
5. Security of the deployment environment
6. Continuous monitoring, etc.

Model

Foundation Model

Fine-Tuned Model

Open Source vs. Closed Source Models

Domain-Specific Models

Model cards

Orchestrated Services

1. Caching Services
2. Security Gateways (LLM Gateways)
3. Deployment Services
4. Monitoring Services
5. Optimization Services
6. Plug-ins for Security
7. Plug-ins for Customization and Integration
8. LLM General Agents

AI Applications

- Direct touchpoint between LLM technology and end-users
- Interface for users to interact with the underlying intelligence of LLMs.
- Facilitate seamless communication
- Task automation across various domains.
- Amplify the benefits or risks associated with LLMs

AI Controls Framework:

- To prioritize the governance and oversight of AI applications.
- Can proactively address the challenges and risks associated with LLM-powered applications
- Should facilitate continuous monitoring and evaluation of AI applications

LLM-Service Lifecycle

Preparation:

- Data collection
- Data curation
- Data storage
- Resource provisioning
- Team and expertise

Development:

- Design
- Training
- Key considerations during development
- Guardrails

Evaluation/Validation

- Evaluation
- Validation/Red Teaming
- Re-evaluation
- Key considerations during evaluation/validation

Deployment:

- Orchestration
- AI Services supply chain
- AI applications

Delivery:

- Operations
- Maintenance
- Continuous monitoring
- Continuous improvement

Service Retirement:

- Archiving
- Data deletion
- Model disposal

LLM Impact Categories

Initial list of high-level impact categories of LLM-related risks:

Loss of Confidentiality	Loss of Integrity	Loss of Availability
Exposed or leaked to unauthorized individuals. E.g. personal data, trade secrets, or other confidential material.	LLM's data or its generated outputs are altered or corrupted. Maliciously or accidentally: incorrect or misleading results.	Disruption to the LLM's operation, preventing users from accessing it when needed. E.g. denial-of-service attacks, system failures, unexpected downtime, excessive billing quotas or computational resources.

This could be expanded to include the new categories of ‘Abuse/Misuse’ and ‘Loss of Privacy” (according to the NIST document AI 100-2 E2023)

LLM Service Threat Categories

1. Model manipulation
2. Data poisoning
3. Sensitive data disclosure
4. Model theft
5. Model Failure/malfunctioning
6. Insecure supply chain
7. Insecure apps/plugins
8. Denial of Service (DoS)
9. Loss of governance/compliance

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